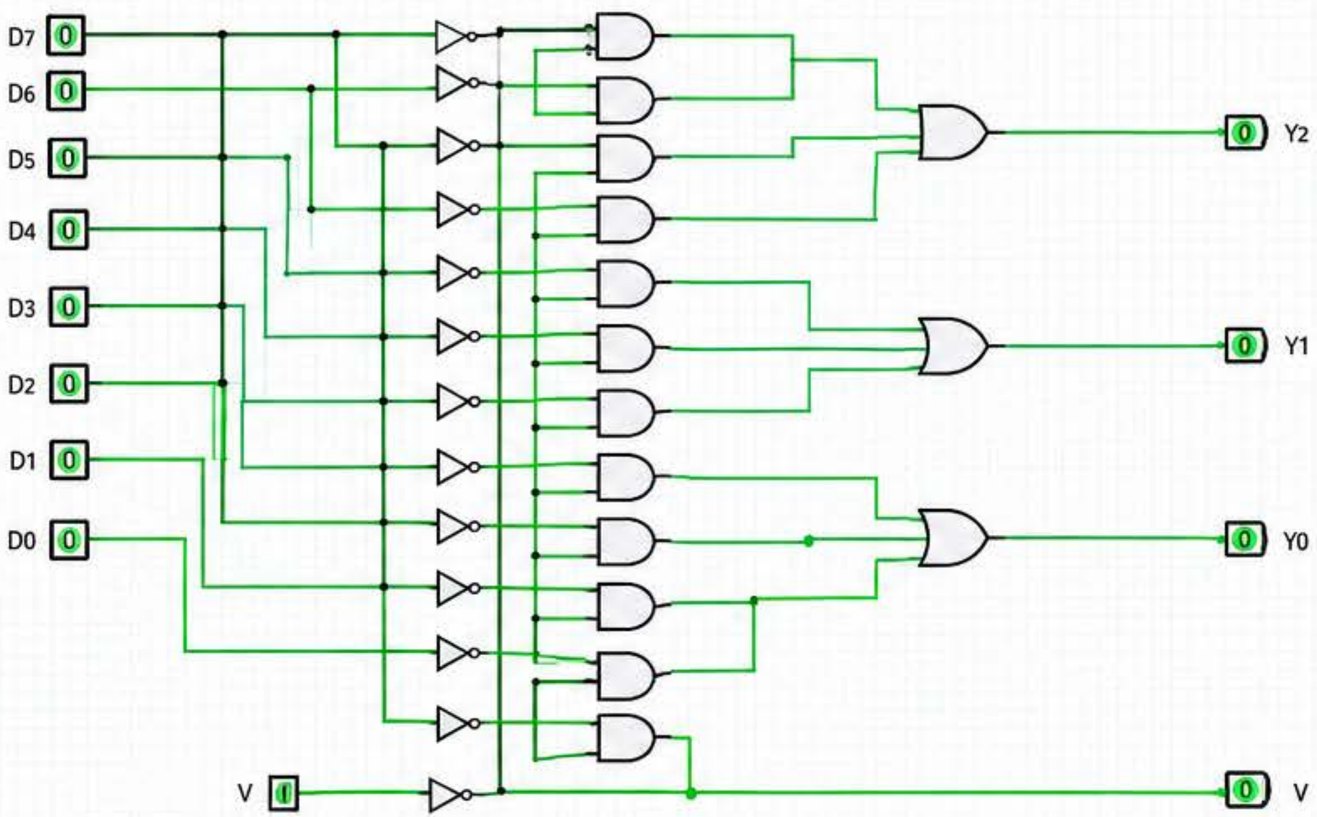
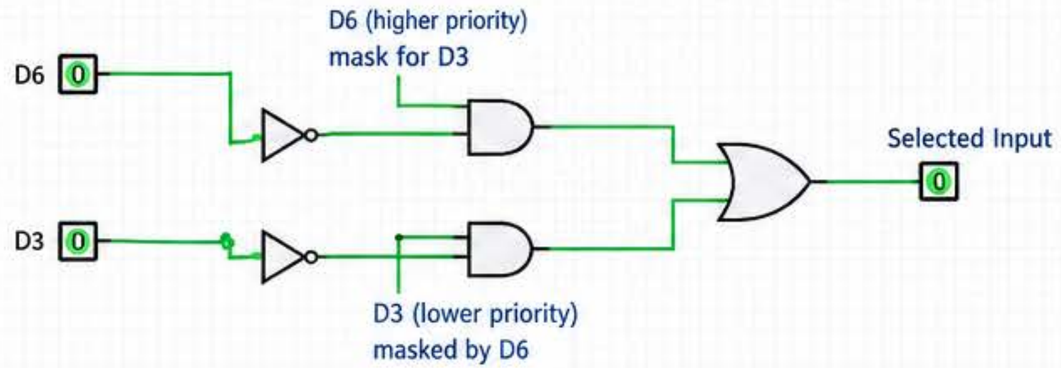
8-to-3 Priority Encoder – Logisim Circuit Diagrams

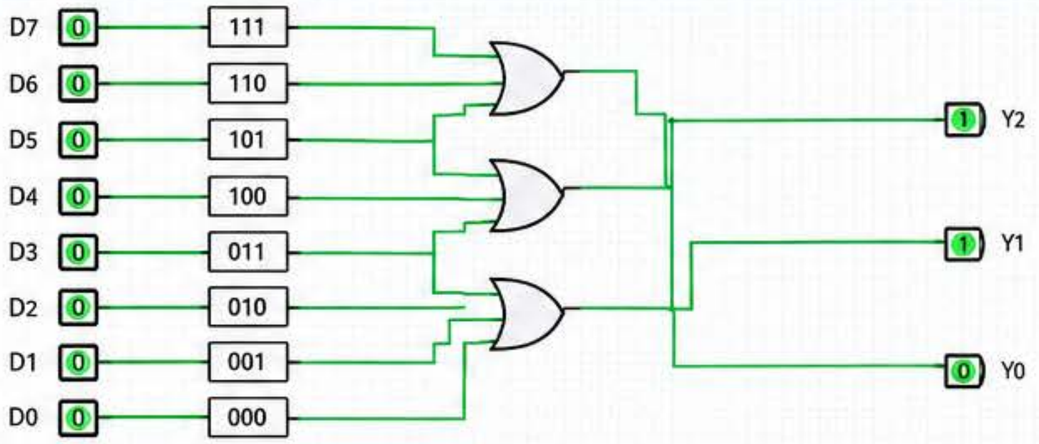
1. Complete Circuit Diagram (8 inputs → 3 outputs)



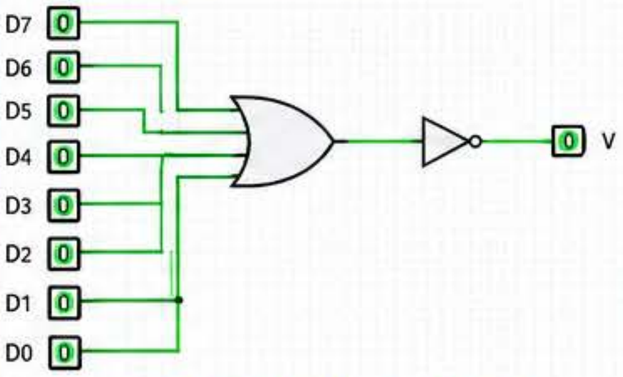
2. Priority Masking Logic (D6 and D3 example)



3. 3-bit Encoding Logic

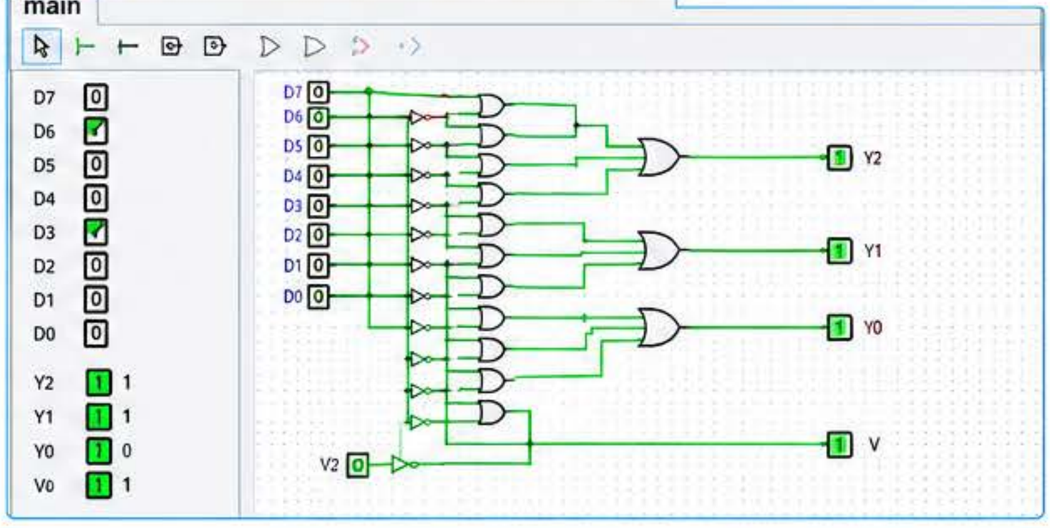


4. Valid Signal (V) Logic



V = 1 when at least one input is HIGH
(Using OR + NOT)

5. Logisim Simulation (Example)



6. Truth Table (Example Cases)

D7	D6	D5	D4	D3	D2	D1	D0	Selected	Y2	Y1	Y0	V
1	0	0	0	0	0	0	1	D7	1	1	1	1
0	1	0	0	0	0	0	0	D6	1	1	0	1
0	0	1	0	0	0	0	0	D5	1	0	1	1
0	0	0	0	0	0	1	0	D3	0	1	1	1
0	0	0	0	0	0	0	1	D0	0	0	0	1
0	0	0	0	0	0	0	0	None	0	0	0	0

Note:

- D7 has the highest priority and D0 the lowest.
- If no input is HIGH, V = 0 and output = 000.